A Lightweight Multiscale Feature Fusion Network for Remote Sensing Object Counting

Jun Yi, *Member, IEEE*, Zhilong Shen, Fan Chen, Yiheng Zhao, Shan Xiao, and Wei Zhou, *Member, IEEE*

***Abstract*— In recent decades, remote sensing object counting has attracted increasing attention from academia and industry due to its potential benefits in urban traffic, public safety, and road planning. However, this issue is becoming a challenge for computer vision because of various technical barriers, such as large-scale variation, complex background interference, and nonuniform density distribution. Recent results show hopeful prospects for object counting using convolutional neural networks (CNNs), but most existing CNN-based methods draw on larger and more complex architectures, which leads to a huge computational and storage burdens, severely limiting their application in real-world scenarios. In this article, a lightweight multiscale feature fusion network for remote sensing object counting, named LMSFFNet, is presented to achieve a better balance between the running speed of the network and the counting accuracy. Specifically, in the encoding process, we select a MobileViT module as the backbone of the network to reduce the numbers of network parameters and computing cost. In return, a cascade structure of the channel–spatial attention mechanisms compensates for the weaker feature extraction ability of the lightweight network. In the decoding process, a lightweight multiscale context fusion module (LMCFM) as a multiscale feature fusion module is developed to solve the problem that the number of parameters increases with the expansion of the object scale when extracting multiscale features. In addition, a lightweight counting scale pooling module (LCSPM) is used to mine the subtle features of the target object. Two kinds of typical object counting experiments, namely, experiments on remote sensing benchmarks (RSOC dataset) and crowd benchmarks (ShanghaiTech, UCF-QNRF, and UCF_CC_50 datasets), show the effectiveness of the proposed method.**

***Index Terms*— Attention mechanism, efficient and lightweight network, multiscale feature fusion, object counting.**

I. Introduction

THE purpose of object counting is to estimate the number of predefined object instances (classified or not) in specific scenes in images and videos [1]. Due to its potential value in urban traffic [2] and public safety [3], it has attracted increasing attention from academia and industry in recent decades. As the application scenarios continue to expand, the objects that need to be counted become various, such as crowds [4], cell microscopy [5], animals [6], and remote-sensing applications [7], [8], [9], [10]. However, it is not easy to accurately estimate the number of objects from congested scenes. Different from crowd counting, challenges to the accuracy of remote sensing object counting include the following: large-scale variation, complex background interference, and nonuniform density distribution [11].

In recent years, a variety of impressive research works on object counting based on convolutional neural networks (CNNs) have emerged along with improved computational capabilities [12], [13], [14]. To extract more features, most state-of-the-art methods actively choose heavy networks or complex multiscale convolutions as the backbone. In addition, some specific architectures, such as multicolumn networks or inception blocks with multiple branches, are often used to capture features of different scales [4]. Inevitably, the above techniques incidentally generate a large number of network parameters and a huge computational burden. In the era of mobile computing and edge computing, it is almost equivalently important to obtain an acceptable accuracy and to achieve fast inference speed with a limited computation budget for edge devices or embedded systems. Therefore, lightweight network architecture has become an effective tool for achieving this goal and a hot topic in the field of computer vision [15]. This ongoing research demonstrates hopeful prospects for realizing efficient object counting using lightweight networks [16]. However, according to previous works, the available networks have various disadvantages that cannot be ignored despite their previous success. To obtain a faster inference speed, most lightweight network architectures use a small number of convolutional layers, which leads to a very limited feature extraction capability of the network. Balancing the relationship between acceptable precision and fast inference speed is always a dilemma for lightweight network designers. In addition, the existing multiscale fusion modules usually use ordinary convolution to extract features of different scales. Although this design can be used to extract and fuse rich scale information, it imposes a large computational burden. Expanding the receptive field of the network with few parameters and low calculation cost to obtain a performance similar to that of the traditional multiscale fusion method is another problem that lightweight network designers need to solve. Apparently, a lightweight network for object counting requires a better balance between the accuracy

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and the interference speed of the network to meet the real-time requirements of various practical applications.

In this article, a lightweight multiscale feature fusion network for remote sensing object counting, named LMSFFNet, is presented for application to practical scenarios. Specifically, in the encoding process, we select a MobileViT [17] module as the network backbone to reduce the numbers of network parameters and computing cost. Furthermore, the cascade structure of the channel attention (CA) mechanism and the spatial attention (SA) mechanism compensates for the weaker feature extraction ability of the lightweight network. In the decoding process, a lightweight multiscale context fusion module (LMCFM) as a multiscale feature fusion module is developed to solve the problem that the numbers of parameters increases with the expansion of the object scale when extracting multiscale features. In addition, the lightweight counting scale pooling module (LCSPM) is used to mine the subtle features of the object.

The main contributions of our work are summarized as follows.

1) We design an efficient and lightweight network (LMSFFNet) based on an encoder–decoder framework for remote sensing object counting, which can realize a better balance between the accuracy and the interference speed of the network to meet the real-time requirements.
2) To extract multiscale high-level feature information, an LMCFM is developed as a multiscale feature fusion module to address the issue that the number of parameters increases with the expansion of the object scale.
3) The LCSPM with few parameters and low calculation cost is proposed for mining the subtle features of the target object. Two kinds of typical object counting experiments, including experiments on remote sensing benchmarks (RSOC dataset [7]) and crowd benchmarks (ShanghaiTech [18], UCF-QNRF [19], and UCF_CC_50 [20] datasets), show the effectiveness of the proposed method.

II. Related Work

A. Object Counting

Early object counting methods, which were used before CNN-based methods become mainstream in this field and include detection-based methods [21] and regression-based methods [22], have substantial limitations in crowded scenes and cannot obtain satisfactory accuracy. In recent years, numerous excellent CNN-based methods [18], [23] have been designed to address the counting task due to the powerful feature representation capabilities of CNNs.

To obtain more abundant feature details, most counting methods actively choose a heavy network or complex multicolumn convolution architecture as the backbone of the network [11]. For instance, the congested scene recognition net (CSRNet) [12] was proposed for interpreting highly congested scenes and performing accurate count estimation, as well as obtaining high-quality density maps. For variable pedestrian or object sizes, the scale pyramid network (SPN) [4] adopts a shared single deep column structure and extracts multiscale information in high layers using a scale pyramid module. In [13], a spatial fully convolutional network (SFCN) was designed to exploit large-scale synthetic data to pretrain a crowd counter. A multicolumn CNN (MCNN) [18] is a simple multicolumn network for solving the problem of scale variation, in which each column is built with different filter kernels.

In the specific object counting, the more challenging remote sensing object counting tasks lie in larger scale variations, more complex disturbances, and more heterogeneous cluster distributions. A variety of impactful works have emerged in remote sensing object detection such as remote-sensing Spatial Adaptation detector (RSADet) [24], embedded graph attention network (EGAT)-long short-term memory mechanism (LSTM) [25], hyper-scale object detection network (HyNet) [26], and scale-aware trident network (SATNet) [27]. Generally, CNN-based object detection and object counting are classified into different task categories due to the large differences in the features of their interest and processing techniques. The object detection, as an extension of object classification, aims only at identifying objects in images, and its task is to detect all the instances of predefined classes and provide their coarse localization in the image by means of axis-aligned boxes. While the density map-based methods, as the current mainstream in object counting, use CNNs to regress a density map from an input image and get the counting result by summing up the map. In fact, the detection methods usually fail when they are used to detect small and dense objects in remote sensing images due to the variations in the appearance and size of the objects. In addition, it is difficult to label small objects with the bounding box that is usually used for the object detection tasks. For these reasons, object counting from remote sensing images is barely studied. Attention scale pyramid deformable network (ASPDNet) [7], as a benchmark method, was designed to accommodate large-scale dataset for remote sensing object counting. PSGDNet [11], which incorporates a pyramidal scale module and a global context module, was proposed to count the accurate number of object instances in remote sensing images. To remedy the same problems, a novel remote sensing multi-scale context feature aggregation (MCFA) method [8] was developed, which contains the adaptive multiscale context aggregation module and the self-context distillation module.

However, in both generic object counting and specific object counting, a complex network architecture is used as the backbone to address the general challenges in the counting field, such as scale variation and background interference. Spectrum-aware feature augmentation network (SFANet) [28], ASPDNet, SPN [4], pyramidal scale and global context guided network (PSGCNet), spatial-/channel-wise attention regression network (SCAR) [29], and context-aware network (CAN) [30] have been demonstrated to be effective on VGG-16. Similarly, SFCN [13] achieved a good counting performance on ResNet-101.

B. Lightweight Networks

To improve the counting efficiency and reduce the computational complexity, scholars have focused on simpler network structures. A trellis encoder–decoder network (TEDnet) [31] was presented for crowd counting, which focuses on generating high-quality density estimation maps. In [32],

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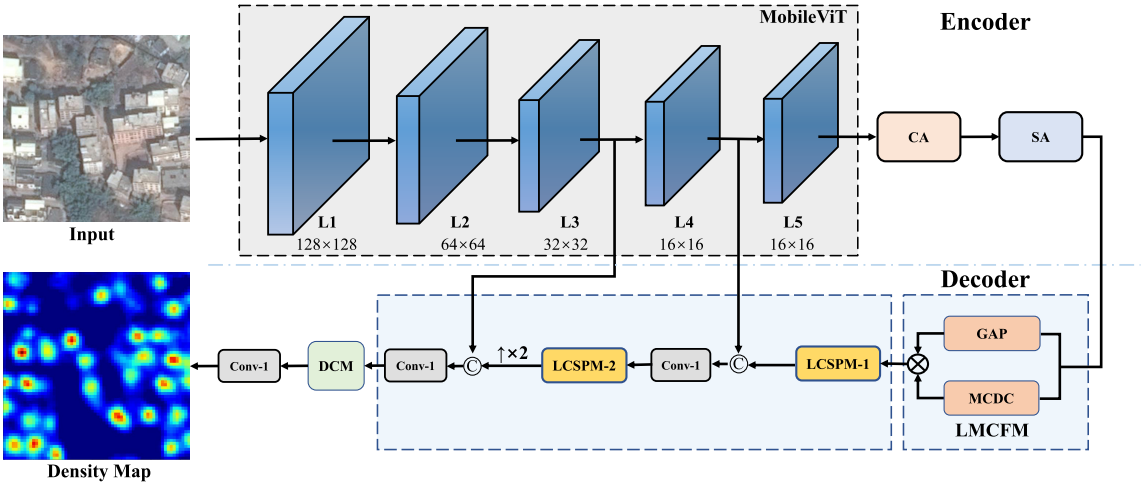


Fig. 1. Overall architecture of the proposed LMSFFNet. "⊗" indicates the multiplication operation. Conv-n represents a standard $n \times n$ convolution. "©" represents the concat operation. "↑[" manifests the upsampling operation. MobileViT is used as the backbone of LMSFFNet to extract feature from an input image. A hybrid attention module (CA + SA) and LMCFM module are designed to enhance the capability of feature extraction. In addition, the LCSPM module is constructed to further segment the object features. The input of LCSPM-1 is the output of LMCFM, and the input of LCSPM-2 is the multiplication of the $L4$ layer and the output of LCSPM-1. DCM and one convolution layer are used to generate the predicted density map.

a compact CNN (CCNN) was designed for crowd counting, which learns a more efficient model with a small number of parameters. Chen et al. [33] introduced an efficient deep learning model structured like a flounder, named FlounderNet, in which a novel interleaved group convolution is applied to eliminate the redundancy of the network. Although these above efforts have partially resolved the contradiction between the accuracy and the effectiveness of counting, most have not considered feature fusion from a lightweight perspective.

In the past several years, lightweight neural architectures have been an active area of interest for striking an optimal balance between the accuracy and the efficiency for object counting [34]. MobileNets [35], which are a class of efficient models for mobile and embedded vision applications, use depthwise separable convolutions to build lightweight deep neural networks. In MobileNetV2 [36], a new mobile architecture, the intermediate dilation layer uses lightweight depthwise convolutions to filter features as a source of nonlinearity. Xu et al. [37] proposed a CityCam-to-Edge cooperative learning framework by integrating multiple city cameras with an edge server to count vehicles more efficiently. Based on MobileNetV2, a computationally efficient encoder–decoder architecture (MobileCount) [16] was proposed to further boost the counting performance without increasing the number of FLOPs. MobileViT [17], a lightweight and general-purpose vision transformer for mobile devices, was presented for learning global representations with transformers.

In this work, LMSFFNet, as an efficient and lightweight network, can realize a better balance between the accuracy and the interference speed of the network to meet the real-time requirements.

III. Proposed Method

A. Overview of the Network Architecture

The architecture of LMSFFNet is shown in Fig. 1, which is mainly composed of a multiscale parallel convolution encoder and decoder.

1) Encoder Module: MobileViT is chosen as the lightweight backbone to extract the image features. Inspired by convolutional block attention module (CBAM) [38], a hybrid attention module is integrated to capture more context information and high-level semantic information. It is difficult to separate the texture features from the background when the objects in the foreground are particularly dense. CA can solve this problem. At the same time, the feature maps at different locations have different density information, so SA is introduced to further encode long-distance dependencies in the spatial dimension, which effectively improves the performance at spatial locations.

2) Decoder Module: LMCFM is designed to capture the multiscale features by building four parallel different receptive fields. There is one LCSPM layer after LMCFM. Then, we fuse the feature map of $L4$ layer and use LCSPM to further enhance the ability to capture object features. In addition, the scale of the feature map is expanded to $L3$ layer through upsampling, and the feature map is fused with the $L3$ layer. The network can better perceive the position information of the target object through the introduction of the dynamic convolutional module (DCM) module. Finally, a 1×1 convolutional layer is used to generate a density map and obtain the final predicted number of objects by summing the values of all the pixels in the density map.

B. Lightweight MultiScale Context Fusion Module

Dilation convolution is often used for the segmentation task, which enables us to obtain high-level multiscale contextual information without increasing the number of parameters or operations. In the object counting task, as a multiscale features extractor, a multicolumn parallel CNN has been proven to be feasible [7]. When the receptive field is expanded, this operation using a multicolumn network does not increase the number of parameters or operations, and its numbers of input and output channels are consistent. However, the disadvantage is also obvious, the larger the size of the feature map, the larger the convolution kernel needed.

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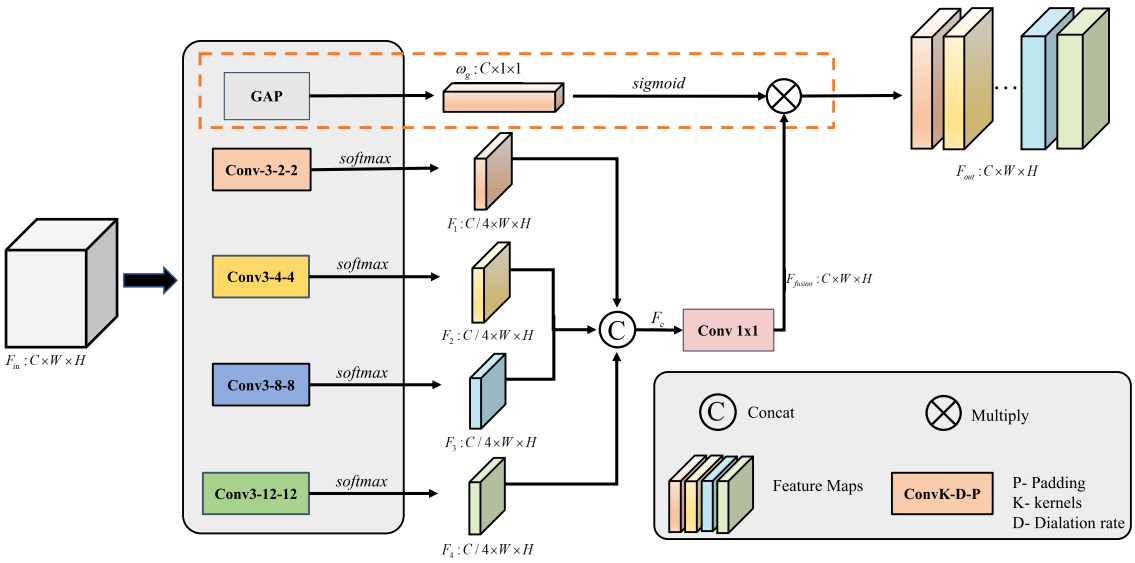


Fig. 2. Details of LMCFM architecture. Multiscale features are captured by four parallel different receptive fields. The parameters of dilation convolutional layers are denoted as "Conv-(kernel size)-(dilation rate)-(padding)." All the convolutional layers maintain the previous size using padding and followed by the softmax layer.

In addition, there is a maximum pooling operation in the encoding phase, which is applied to retain the features of the input while simultaneously reducing the amount of data. However, there are also two shortcomings. First, the size of the feature map is reduced to 1/8 of the original size when the number of channels of the output feature map increases. With the reduction in the size of the feature map, it becomes more difficult for the network to capture the features of small objects. At the same time, the low parameter model has various limitations in feature extraction, resulting in the model neglecting small objects. Second, the maximum pooling operation has local invariance, which makes it more suitable for classification tasks and not conducive to object location.

Algorithm 1 LMCFM

Input: Feature map $F_{in} \in \mathbb{R}^{C\times H\times W}$.
Output: Multi-Scale feature map F_{out}.

```
Define convolution layer D_n, dilation rate={2, 4, 8, 12}.
for (D_i) in (D_n) do
    Apply D_i block to feature map F_in.
    Obtain dilated feature map F_i.
    Apply softmax function to corresponding pixels of the feature map.
end for
Calculate fusion feature map F_fusion by 1 and 2, and the weight value ω_g by 3.
Calculate the final merged feature F_out by 4.
return F_out.
```

Therefore, in response to the above-mentioned questions, the LMCFM is designed to extract multiscale information and high-level features and reduce the numbers of parameters and operations by four parallel different receptive fields. In addition, a global average pool (GAP) layer is used to calculate an average value for all the pixels of the characteristic map input to each channel, and then the calculation result is input into the softmax activation function to help the network prevent the LMCFM module from overfitting without increasing the parameter amount. The architecture is shown in Fig. 2.

In LMCFM, the dilation rates are set to 2, 4, 8, and 12, while the number of output channels is reduced to 1/4 of the original number. Then, four channels are spliced to restore the original number of channels. Finally, the dimension is reduced through convolution. The first task of LMCFM is multiscale information extraction. The convolutional network contains four branches. An explanation is given as follows. $\mathcal{F}_2$: $F_{\text{in}} \rightarrow F_1$. $\mathcal{F}_1$. $\mathcal{F}_2$ represents the multiscale feature extracted by the convolutional layer with a dilation rate of 2 in the first dilated convolution. $\mathcal{F}_4$: $F_{\text{in}} \rightarrow F_2$. $\mathcal{F}_4$ represents the multiscale feature extracted by the convolutional layer with a dilation rate of 4 in the second dilated convolution. $\mathcal{F}_8$: $F_{\text{in}} \rightarrow F_3$. $\mathcal{F}_8$ represents the multiscale feature extracted by the convolutional layer with a dilation rate of 8 in the third dilated convolution. $\mathcal{F}_{12}$: $F_{\text{in}} \rightarrow F_4$. $\mathcal{F}_{12}$ represents the multiscale feature extracted by the convolutional layer with a dilation rate of 12 in the fourth dilated convolution. Different dilation rates of dilated convolution result in different receptive fields for handling multiscale object information. Then, F_c is concatenated with F_2, F_4, F_8, and F_{12}, which are formulated as follows:

$$F_c = \text{Concat}(F_2, F_4, F_8, F_{12}). \tag{1}$$

The fused features F_c are fed into a 1×1 convolution, and the sigmoid nonlinear activation F_{fusion} is used to generate the confidence map with background removed. Due to the limited feature extraction capability of lightweight architecture, there is still unavoidable background noise in the feature map input to LMCFM module. Specifically, a $C \times 1 \times 1$ vector ω_g is generated by a GAP layer and sigmoid activation function

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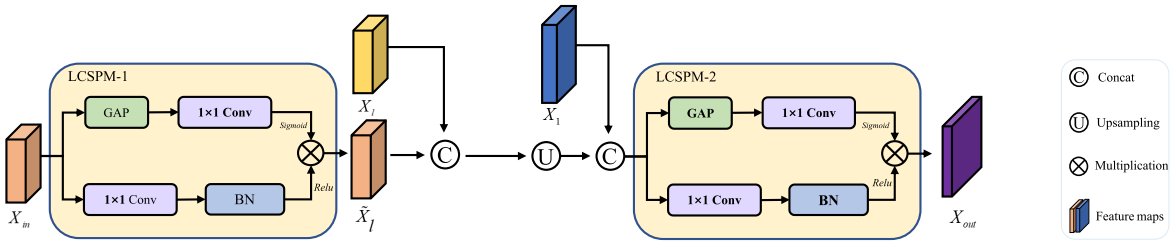


Fig. 3. Details of LCSPM architecture. The output of LMCFM (F_{out}) is used as the input of LCSPM (X_{in}). X_{out} is the merged feature map. $\tilde{X}$ represents the output of LCSPM. l represents the lth layer of the backbone structure, and X_l is the lth feature map captured from the backbone structure with $l \in \{1, 2, 3, 4, 5\}$. The concat operation is performed on X_l and $\tilde{X}$. An upsampling operation has to be performed if the scales of X_l and $\tilde{X}$ are different.

Algorithm 2 Lightweight Counting Scale Pooling Module (LCSPM)

Input: The feature map X_{in} output by LMCFM.
Output: Merged feature map X_{out}.
1: Apply GAP operation on X_{in}.
2: Obtain a $C \times 1 \times 1$ vector ω_p by 5.
3: Calculate the weight ω_c by 6.
4: Apply convolution operation on X_{in} and obtain convolution result X_c.
5: Apply batch normalization (BN) and rectified linear unit (Relu) activation function on X_c and obation X_b.
6: Calculate X_{out} by $\omega_c \otimes X_b$.
7: **return** X_{out}.

and multiplied with F_c. In this way, the number of parameters of the network structure is reduced, and the local and global information are fully combined so that the network can better extract the key information of the target object. The specific formula is as follows:

$$F_{\text{fusion}} = \text{Conv}(F_c) \tag{2}$$

$$\omega_g = \text{Sigmoid}(\text{GAP}(F_{\text{in}})) \tag{3}$$

where F_{in} represents the input feature map, and $Conv$ is the convolution operation. A weighting ω_g is obtained through global average pooling and then multiplied by F_{in}, which formulated as follows:

$$F_{\text{out}} = \omega_g \otimes F_{\text{fusion}} \tag{4}$$

where "$\otimes$" indicates the multiplication operation.

Using this approach, all the layers contain the same number of channels but use different dilation rates (2, 4, 8, and 12) to capture multiscale information. Consequently, four different receptive fields are used to build a pyramid multiscale structure, which can extract rich feature information. Then, the global weighting coefficient is multiplied by a multiscale feature map to obtain the global information while reducing the numbers of parameters and computations without changing the scales.

C. Lightweight Counting Scale Pooling Module (LCSPM)

After feature extraction of the LMCFM module, the feature information of the objects needs to be further segmented from the feature map and counted. The scale of the feature map introduced from LMCFM has been reduced by 1/64. Inspired by MobileNetV3 [39], the lightweight counting spatial pooling module is constructed to efficiently segment the semantic information of the dense object and fuse local and global information. A structure diagram is shown in Fig. 3. It contains two branches. An explanation is given as follows. In the upper part, the feature map X_{in} is passed through a GAP layer, and a $C \times 1 \times 1$ vector ω_p is obtained. The specific formula is as follows:

$$\omega_p = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j) \tag{5}$$

where H and W are the height and width of the feature map, respectively. x_c is the cth channel of the feature map X_{in}, and $c \in \{1, 2, \ldots, C\}$. Then ω_p is passed through a 1×1 convolutional layer to reduce the number of channels and parameters. Finally, ω_c represents the channel weight generated by a nonlinear activation function as follows:

$$\omega_c = f\big(\text{Conv}\big(\omega_p\big)\big) \tag{6}$$

where f represents the sigmoid activation function, and $Conv$ means a 1×1 convolutional layer.

On the other branches, $X_{\text{in}} \rightarrow X_c$ represents the number of channels reduced by 1×1 convolution layer. Then, the convolution result is normalized by BN operation and activated by Relu function, formulated as $X_b \rightarrow \text{BN}(X_c)$. Finally, the two branches are multiplied together and formulated as $\tilde{X} : \omega_c \otimes X_b$ where $\tilde{X}$ represents the result of the channel coefficient ω_c multiplied by X_b. Immediately afterward, the concat operation is performed on X_l and $\tilde{X}$ where X_l is the feature map of the lth layer in backbone, and its formula is shown as follows:

$$\hat{X}_l = \text{Concat}\big(X_l, \tilde{X}_l\big). \tag{7}$$

Meanwhile, since LCSPM does not change the scale of the feature map, it is necessary to expand the scale of the feature map by upsampling. Compared with using a large convolution kernel to reduce the number of channels, LMSFFNet not only achieves a low number of parameters but also fully integrates global and local information, enabling the network to better segment the location information of the object and improve the accuracy of counting. The pseudocode for LCSPM is presented as Algorithm 2.

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TABLE I
REMOTE SENSING OBJECT COUNTING DATASETS USED FOR EVALUATION

Datasets		Platform	Images	Training/test	Average Resolution	Annotation Format	Count Statistics			
							Total	Min	Average	Max
RSOC [7]	Building	Satellite	2468	1205/1263	512×512	center point	76,215	15	30.88	142
	Small-vehicle	Satellite	280	222/58	2473×2339	oriented bounding box	148,838	17	531.56	8531
	Large-vehicle	Satellite	172	108/64	1552×1573	oriented bounding box	16,594	12	96.48	1336
	Ship	Satellite	137	97/40	2558×668	oriented bounding box	44,892	50	327.68	1661
CARPK [40]		Drone	1448	989/459	1280×720	bounding box	89,777	1	62	188
PUCPR+ [40]		Camera	125	100/25	1280×720	bounding box	16,915	0	135	331
DroneCrowd [41]		Drone	33,600	24,600/9,000	1920×1080	bounding box	4,864,280	25	144.8	455

TABLE II
CROWD COUNTING DATASETS USED FOR EVALUATION

Datasets	Avg.	Max.	Total	Numbers of Image	Average Resolution
ShanghaiTech A [18]	501	3,139	241,667	482	589 × 868
ShanghaiTech B [18]	123	578	88,488	716	768 × 1024
UCF_CC_50 [20]	1,279	4,633	63,974	50	2101 × 2888
UCF-QNRF [19]	815	12,865	1,251,642	1,525	2013 × 2902

D. Ground-Truth Generation

The ground truth (GT) is generated based on previous works. A object instance can be expressed as $\delta(x - x_i)$ at the pixel $\S_i$. Therefore, for an image with N objects, it can be expressed in the following form:

$$H(x) = \sum_{i=1}^{N} \delta(x - x_i). \tag{8}$$

To transform $H(x)$ to a continuous density function, the usual method is to widen a field of point coordinates through a Gaussian kernel function, which are formulated as follows:

$$F(x) = \sum_{i=1}^{N} \delta(x - x_i) * G_{\sigma_i}(x) \tag{9}$$

where $G_{\sigma_i}(x)$ is the Gaussian kernel function, and σ_i represents the standard deviation.

E. Loss Function

The Euclidean distance is adopted as a loss function to evaluate the difference between the predicted density maps and the GT. The loss function $L(\Theta)$ is defined as follows:

$$L(\Theta) = \frac{1}{2M} \sum_{i=1}^{M} \left\| F(C_i; \Theta) - F_i^{\mathrm{GT}} \right\|_2^2 \tag{10}$$

where M represents the number of pixels in the density maps, C_i represents the input image, θ represents the training parameters, and $F(C_i; \Theta)$ and F_i^{GT} represent the estimated number and the GT, respectively.

The counting loss (L_c) is introduced in our work; it directly measures the difference between the predicted number and the GT. Through continuous training and learning of the network, the count difference is minimized. L_c is defined as follows:

$$L_c = \frac{1}{N} \sum_{i=1}^{N} \frac{C^p(i) - C^g(i)}{(C^g(i) + \gamma)^2} \tag{11}$$

where N is the number of samples, $C^p(i)$ represents the predicted value, $C^g(i)$ represents the actual value, and i represents the ith training sample. γ is a super parameter to prevent the denominator from being 0 and set it to 1e−4. The final loss L is defined as follows:

$$L = L(\Theta) + \lambda L_c. \tag{12}$$

In this work, λ is 1e−2.

IV. EXPERIMENTS

In this section, the datasets, implementation details, and evaluation strategy are first introduced. Then, we evaluate our method on the state-of-the-art remote sensing datasets and dense crowd datasets to demonstrate the superiority performance of LMSFFNet. Furthermore, we also conduct ablation experiments on our main innovations.

A. Datasets and Evaluation Metrics

Four remote sensing datasets (RSOC [7], CARPK [40], PUCPR+ [40], and DroneCrowd [40]) are used to demonstrate the superior performance of our proposed approach. The datasets are described in detail in Table I. Furthermore, we also conducted experiments on dense crowd datasets (ShanghaiTech Part_ A/B [18], UCF-QNRF [19] and UCF_CC_50 [20]) to verify the generalization ability and robustness of the model, as described in detail in Table II.

By following existing works, both the absolute error (MAE) and the mean-squared error (RMSE) are chosen as the performance indicators of the evaluation method, which are formulated as follows:

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=0}^{N} |y_i - \hat{y}_i| \tag{13}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum_{i=0}^{N} |y_i - \hat{y}_i|^2} \tag{14}$$

where N represents the number of samples, and y_i and $\hat{y}_i$ represent the GT and the predicted value, respectively. In addition, the peak signal-to-noise ratio (PSNR) and structural similarity

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TABLE III

COMPARISON OF OUR PROPOSED METHOD WITH STATE-OF-THE-ART METHODS IN TERMS OF MAE, RMSE, PARAMETERS, AND FLOPS ON THE RSOC DATASET; THE UNIT OF FLOPS IS GIGA (G), THE FLOPS ON 1024 × 768 INPUTS ARE REPORTED

Methods	Backbone	Years	Params(M)	GFlops	RSOC_Building		RSOC_Small-vehicle		RSOC_Large-vehicle		RSOC_Ship	
					MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
CSRNet [12]	VGG-16	2018 CVPR	16.26	325.3	8.00	11.78	443.72	1252.22	34.10	46.42	240.01	394.81
SFCN [13]	Resnet-101	2019 CVPR	-	-	8.94	12.87	440.70	1248.77	33.93	49.74	240.16	394.81
SPN [4]	VGG-16	2019 WACV	-	-	7.74	11.48	445.16	1252.92	36.21	50.65	241.43	392.88
SCAR [29]	VGG-16	2019 NC	-	-	26.90	31.35	497.22	1276.65	62.78	79.64	302.37	436.92
CAN [30]	VGG-16	2019 CVPR	18.18	-	9.12	13.38	457.36	1260.39	34.56	49.63	282.69	423.44
SFANet [28]	VGG-16	2019 CVPR	-	-	8.18	11.75	435.29	1284.15	29.04	47.01	201.61	332.87
ASPDNet [7]	VGG-16	2021 TGRS	22.702	465.75	7.59	10.66	433.23	1238.61	18.76	31.06	193.83	318.95
MCFA [8]	VGG-16	2021 TGRS	-	-	7.93	11.82	238.46	625.90	12.94	20.25	50.45	65.24
PSGCNet [11]	VGG-19	2022 TGRS	27.51	385.79	7.54	10.52	157.55	245.31	**11.00**	**17.65**	74.91	112.11
MCNN [18]	FS	2016 CVPR	**0.13**	21.17	13.65	16.56	488.65	1317.44	36.56	55.55	263.91	412.30
CMTL [42]	-	2017 AVSS	2.454	95.55	12.78	15.99	490.53	1321.11	61.02	78.25	251.17	403.07
SANet [43]	-	2018 ECCV	1.39	71.45	29.01	32.96	497.22	1276.66	62.78	79.65	302.37	436.91
MobileCount [16]	MobileNetV2	2020	3.40	6.15	7.72	11.90	316.02	598.58	18.5	30.4	73.2	100.2
LMSFFNet-XXS	MobileViT	-	2.49	**6.04**	7.12	10.35	198.14	494.33	16.27	30.13	61.61	85.03
LMSFFNet-XS	MobileViT	-	4.58	14.9	6.52	**9.67**	**141.7**	**273**	12.74	27.13	**49.47**	**70.0**
LMSFFNet-S	MobileViT	-	9.49	41.3	**6.33**	9.38	205.03	503.47	13.15	29.47	57.15	80.0

index measure (SSIM) in each image are applied to evaluate the quality of the density maps.

B. Implementation Details

LMSFFNet is implemented by the PyTorch toolbox, and all the experiments are conducted on one NVIDIA TITAN RTX GPU. The Adam optimizer is used as the optimizer, and the initial learning rate is 1e−4 without learning rate decay. Hyperparameters λ are set to 0.01. For the building subset, we adopt a batch size of 8 and set a batch size of 1 for the others. The training process can converge within 300 epochs. Small-vehicle, large-vehicle, and ship images are uniformly set to the size of 1024 × 768. Random cropping and horizontal flipping are applied for augmentation to improve training and avoid overfitting. Specifically, the crop size is 256 × 256 for the RSOC_Building, ShanghaiTech Part_A, and UCF_CC_50 datasets, and 512 × 512 for the others.

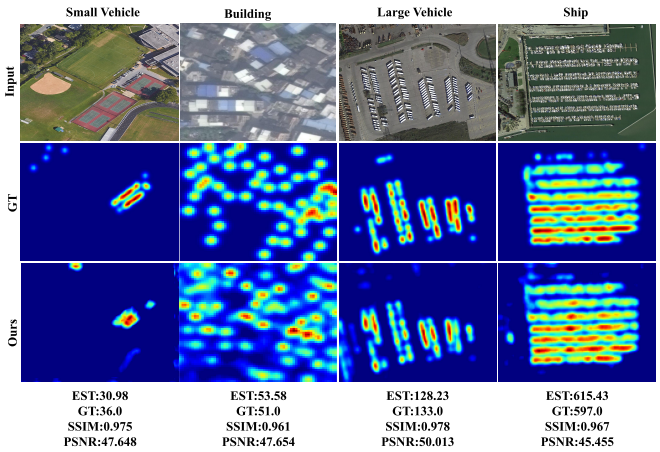


Fig. 4. Visualization results of LMSFFNet on the RSOC dataset. The first row represents the original images, the second row represents the GT, and the third row represents the predictions of LMSFFNet.

C. Remote Sensing Object Counting

1) Performance on RSOC: The RSOC datasets include buildings, small vehicles, large vehicles, and ships. Among them, different categories of images have different resolutions and object sizes. For instance, the resolution of small vehicle images is 2473 × 2339, and there are an average of 531 object instances per image. The tested methods are divided into two groups. Those in the first group have complex network structures, with large numbers of parameters. Those in the second group have lightweight network structures, with fewer parameters. We compare the state-of-the-art remote counting approaches with our proposed LMSFFNet, for which three different sizes (S: small, XS: extra small, and XXS: extra extra small. More details about their parameters are shown in the literature [17].) are considered, and the results are shown in Table III. The compared methods are divided into two groups. Those in the first group have complex network structures, with large numbers of parameters. Those in the second group have lightweight network structures, with fewer parameters. Our proposed approach shows the lowest MAE and RMSE values on the building, ship, and small vehicle datasets, as listed in Table III. Compared with the state-of-the-art method PSGCNet [11], taking the small vehicle and building datasets as examples, LMSFFNet-XS significantly improves the MAE from 157.55 to 141.7 and the MSE from 74.91 to 49.47. Although the MAE on large vehicles is lower than that of PSGCNet, our model has fewer parameters and far fewer GFLOPs in the first group. Compared with the second group, our three models achieve better results than MCNN, cascaded multi-task learning (CMTL), scale aggregation network (SANet), and MobileCount in all the categories. It is enough to prove that our network can achieve better performance in large-scale object counting scenes using a low-parameter network structure. Meanwhile, LMSFFNet achieves better SSIM and PSNR values in density map generation, as shown in Fig. 4.

2) Performance on CARPK and PUCPR+: CARPK is a large-scale UAV vehicle counting dataset. As shown in Table IV, our proposed method is compared with not only the state-of-the-art density map estimation-based methods (MCNN [18], CSRNet [12], BL [23], and PSGCNet [11]) but also the detection-based methods such as YOLO [21], Faster

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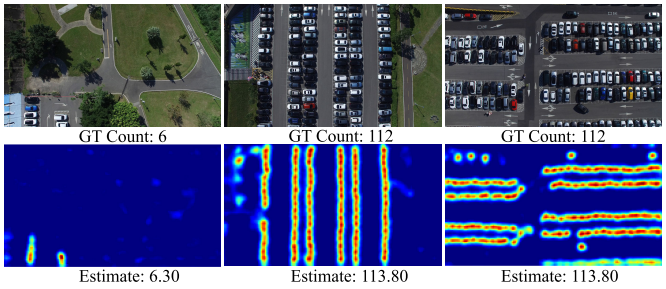


Fig. 5. Visualization results on the CARPK dataset. (Top) Original image and the GT counts. (Bottom) Density maps generated by our proposed method and the estimated counts.

TABLE IV
PERFORMANCE COMPARISON ON THE CARPK [40] AND PUCPR+ [40] DATASET. * INDICATES THAT THE METHOD HAS BEEN FINE-TUNED ON PUCPR+ DATASET

Methods	CAPRK		PUCPR+	
	MAE	RMSE	MAE	RMSE
YOLO [21]	102.89	110.02	156.72	200.54
*YOLO [21]	48.89	57.55	156.00	200.42
Faster RCNN [44]	103.48	110.64	156.76	200.59
*Faster RCNN [44]	24.32	37.62	39.88	47.67
One-Look Regression [22]	59.46	66.84	21.88	36.73
SSD [45]	37.33	42.32	119.24	132.22
YOLO9000 [46]	38.59	43.18	97.96	133.25
LPN [40]	23.80	36.79	22.76	34.46
RetinaNet [47]	16.62	22.30	24.58	33.12
LEP [48]	51.83	-	15.17	-
MCNN [18]	39.10	43.30	21.86	29.53
CSRNet [12]	11.48	13.32	8.65	10.24
BL [23]	9.58	11.38	6.54	8.13
ASPDNet [7]	7.81	10.16	-	-
PSGCNet [11]	8.15	10.46	5.24	7.36
LMSFFNet-XS	**7.05**	**9.03**	**4.49**	**6.21**

RCNN [44], layout proposal network (LPN) [40], single shot multibox detector (SSD) [45], and regression-based methods (one-look regression [22]). The results show that our method achieves the lowest MAE and MSE values on CAPRK and PUCPR+. A visualization of LMSFFNet is shown in Fig. 5.

3) Comparisons on the DroneCrowd Dataset: We also evaluate our approach on a more challenging dataset, DroneCrowd, which contains 112 video clips with 33 600 high-resolution frames (1920 × 1080) captured in 70 different scenarios. The proposed method achieves comparable results in the large, sunny, cloudy, and crowded scenarios and behaves ordinarily in the night scenario, as shown in Table V. Even so, the proposed LMSFFNet achieves better results than DM-Count [56], CAN and PSGCNet [11] in the night scenario. Moreover, although CMTL [42] has fewer parameters than our method, our MAE is 57.6% lower overall, and our LMSFFNet also outperforms CMTL in other categories, especially in crowded areas. In particular, space-time neighbor-aware network (STNNet) [41] achieves the lowest MAE and RMSE on the whole dataset. The proposed method also achieves comparable results on the large, sunny, and crowd scenarios with few parameters. Fig. 6 shows the counting results for the cloudy, night, sparse, and sunny scenarios. They indicate that the proposed LMSFFNet also achieves comparatively excellent performance and even outperforms the state-of-the-art methods in some challenging scenarios.

D. Comparisons on Dense Crowd Counting Datasets

We compare LMSFFNet with various advanced methods on ShanghaiTech PartA/B, UCF-QNRF, and UCF-CC-50, as shown in Table VI, and the specific details are shown in Table II. On ShanghaiTech, Our method performs better than CP-CNN, Switch-CNN, CSRNet, and SUA-Fully with a low number of parameters. Likewise, our method achieves the lowest MAE and MSE in the second group. On the UCF-QNRF datasets, LMSFFNet achieves the best result in the second group. Compared with the first group, the proposed approach (MAE:112.8) performs better than CSRNet and SUA-Fully, and LMSFFNet has fewer parameters than all of them. On the UCF-CC-50 datasets, the counting performance of LMSFFNet is superior than those of all other methods except UEPNet in the first group. In the second group, the proposed method again achieves superior performance because the LMCFM module can better aggregate multiscale features. A visualization of the results is shown in Fig. 6. The first column represents ShanghaiTech A, the second column represents ShanghaiTech B, the third column represents UCF-QNRF, and the last column represents UCF-CC-50.

E. Ablation Studies

1) Ablation Experiment on the Backbone: To evaluate the effectiveness of the backbone of LMSFFNet, we conduct an ablation experiment on the RSOC dataset. The ablation experiment is performed on ResNet-50 [66], VGGNet-16 [8], MobileNetV2 [16], and MobileViT (LMSFFNet), as shown in Table IX. Compared with the above backbone on the RSOC dataset, the same conclusion is obtained.

2) Ablation Experiment on the Attention Module: In this article, we test the effects of three types of attention mechanisms, as shown in Table VIII. The result shows that the cascade structure of the CA and SA mechanisms offers the best performance.

3) Ablation Experiment on Dilation Convolution of LMCFM: To explain the influence of different dilated convolution modules to extract fusion features of various scales, a special ablation experiment is carried out. The experimental results are listed in the first four rows in Table VII. The overall MAE and MSE of the network are the highest when the number of dilation convolutions is 1. As the number of convolutions increases, the MAE and MSE gradually declined. The network achieves the best results when the number of dilation convolutions is 4, especially in the RSOC_Small-Vehicle subset. But when the number of convolutions is 6 (rate = 2, 4, 8, 12, 16, 20), two metrics of MAE and MSE do not achieve the better performances as expected. It indicates that the network has a strong ability to capture long-range information and multiscale information only for the RSOC_Building distributed uniformly and densely. However, the long-distance information extracted is more background information for small-vehicle, ship, and large-vehicle datasets, which leads to poor object counting performances.

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TABLE V

PERFORMANCE COMPARISON ON THE DRONECROWD DATASET [41]

Methods	Params(M)	Speed(FPS)	Overall		Large		Small		Cloudy		Sunny		Night		Crowded		Sparse	
			MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
CMTL [42]	**2.45**	2.31	56.37	65.9	53.5	63.2	61.5	69.7	59.5	66.9	56.6	67.8	48.2	58.3	81.6	88.7	42.2	47.9
MSCNN [49]	-	1.76	58.0	75.2	58.4	77.9	57.5	71.1	64.5	85.8	53.8	65.5	46.8	57.3	91.4	106.4	38.7	48.8
LCFCN [50]	-	3.08	136.9	150.6	126.3	140.3	152.8	164.8	147.1	160.3	137.1	151.7	105.6	113.8	208.5	211.1	95.4	110.0
SwitchCNN [51]	15.3	0.01	66.5	77.8	61.5	74.2	74.0	83.0	56..0	63.4	69.0	80.9	92.8	105.8	67.7	79.8	65.7	76.7
ACSCP [52]	5.1	1.58	48.1	60.2	57.0	70.6	34.8	39.7	42.5	46.4	37.3	44.3	86.6	106.6	36.0	41.9	55.1	68.5
AMDCN [53]	-	0.16	165.6	167.7	166.7	168.9	163.8	165.9	160.5	162.3	174.8	171.1	162.3	164.3	165.5	167.7	165.6	167.8
StackPooling [54]	-	0.73	68.8	77.2	68.7	77.1	68.8	77.3	66.5	75.9	74.0	83.4	65.2	67.4	95.7	101.1	53.1	59.1
DA-Net [55]	-	2.52	36.5	47.3	41.5	54.7	28.9	33.1	45.4	58.6	26.5	31.3	29.5	34.0	56.5	68.36	24.9	28.7
CSRNet [12]	16.26	3.92	19.8	25.6	17.8	25.4	22.9	25.8	12.8	16.6	19.1	22.5	42.3	45.8	20.2	24.0	19.6	26.5
CAN [30]	18.18	7.12	22.1	33.4	18.9	26.7	26.9	41.5	**11.2**	**14.9**	14.8	17.5	69.4	73.6	**14.4**	**17.9**	26.6	39.7
DM-Count [56]	21.5	**10.04**	18.4	27.0	19.2	19.6	17.2	22.4	11.4	16.3	**12.6**	**15.2**	51.1	55.7	17.6	21.8	18.9	29.6
STNNet [41]	-	3.41	**15.8**	**18.7**	**16.0**	**18.4**	**15.6**	**19.2**	14.1	17.2	19.9	22.5	**12.9**	**14.4**	18.5	21.6	**14.3**	**16.9**
PSGCNet [11]	27.51	6.79	24.7	31.9	24.5	32.8	24.0	31.3	18.8	21.6	15.4	18.6	47.5	54.5	18.2	21.2	18.6	21.2
LMSFFNet-XS	4.58	5.75	23.85	30.69	17.52	21.67	27.13	34.19	20.19	25.28	23.07	28.05	36.4	46.5	20.91	26.83	25.55	38.6

TABLE VI

PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE CROWED DATASET

Methods	Venue	Params(M)	ShanghaiTech A		ShanghaiTech B		UCF-QNRF		UCF-CC-50	
			MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
CP-CNN [57]	ICCV17	68.40	73.6	106.4	20.1	30.1	-	-	295.8	320.9
Switch-CNN [51]	CVPR17	**15.30**	90.4	135.0	21.6	33.4	-	-	318.1	439.2
CSRNet [12]	CVPR18	16.26	68.2	115.0	10.6	16.0	135.4	207.4	266.1	397.5
BL [23]	ICCV19	21.50	61.5	103.2	7.5	12.6	87.7	158.1	229.3	308.2
UOT [58]	AAAI21	21.50	58.1	95.9	6.5	10.2	83.3	**142.3**	-	-
SUA-Fully [59]	ICCV21	15.85	66.9	125.6	12.3	17.9	119.5	213.3	-	-
UEPNet [60]	ICCV21	26.21	54.6	91.2	6.4	10.9	-	-	**81.1**	**131.7**
P2PNet [61]	ICCV21	18.34	**52.7**	**85.1**	**6.3**	**9.9**	85.3	154.5	172.7	256.1
DKPNet [62]	ICCV21	30.63	55.6	91.0	6.6	10.9	**81.4**	147.2	-	-
GL [62]	CVPR21	21.51	61.3	95.4	7.3	11.7	84.3	147.5	-	-
MCNN [23]	CVPR16	0.13	110.2	173.2	26.4	41.3	277.0	426.0	377.6	509.1
CMTL [42]	AVSS17	2.47	101.3	152.4	20.0	31.1	252.0	514.0	322.8	397.9
ACSCP [52]	CVPR18	5.10	75.7	102.7	17.2	27.4	-	-	291.0	404.6
TDF-CNN [63]	AAAI18	0.13	97.5	145.1	20.7	32.8	-	-	354.7	491.4
SANet [43]	ECCV18	0.91	75.3	122.2	10.5	17.9	152.6	247.3	258.4	334.9
PCC-Net [64]	TCSVT19	0.55	**73.5**	**102.7**	11.0	19.0	148.7	247.3	240.0	315.0
LCNet [65]	ICIP19	0.86	93.3	149.0	15.3	25.2	-	-	326.7	430.6
C-CNN [32]	ICASSP20	**0.073**	88.1	141.7	14.9	22.1	-	-	-	-
LMSFFNet-XS	-	4.58	85.85	139.9	**9.2**	**15.1**	**112.8**	**201.6**	**105.7**	**120.3**

TABLE VII

ABLATION STUDY ON LMCFM AND LCSPM

Methods	Building		Ship		Large-Vehicle		Small-Vehicle	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
LMCFM w/ DC 1	7.14	10.4	67.21	87.08	16.84	42.3	190.03	354.6
LMCFM w/ DC 2	7.17	9.95	57.44	80.32	16.14	42.53	180.08	344.82
LMCFM w/ DC 3	6.98	9.67	52.75	73.55	20.74	60.07	230.42	469.34
LMCFM w/ DC 6	6.82	10.19	58.63	79.66	21.5	40.17	187.01	384.69
LCSPM*0	8.98	12.42	72.29	101.34	21.96	40.02	271.44	647.83
LCSPM*1	7.93	11.82	69.8	100.09	16.25	35.83	233.07	613.43
LCSPM*3	28.14	61.47	241.48	319.95	59.08	109.13	971.05	2780.32
LMCFM w/ DC 4 and LCSPM*2	**6.52**	**9.67**	**49.47**	**70.0**	**12.74**	**27.13**	**141.7**	**273**

Note: DC represents the dilation convolution network. LMCFM w/ DC n indicates that the number of DCs in LMCFM is n, while LCSPM*n denotes that the number of LCSPM in LMSFFNet is n.

TABLE VIII

ATTENTION MECHANISM ABLATION STUDIES ON THE RSOC SMALL-VEHICLE DATASET

Methods	RSOC_Small-Vehicle	
	MAE	RMSE
Baseline	250.18	537.43
Baseline+SE	238.41	559.73
Baseline+Coordinate	212.0	445.0
Baseline+CA+SA	**141.7**	**273.0**

4) Effect of the Hyperparameter λ: To verify the effectiveness of the loss function, we conduct experiments under the condition of different λ. As can be observed from Table X, when $\lambda = 0.01$, we can obtain the best performance.

5) Ablation Experiment on LCSPM: To study the influence of LCSPM, we design an ablation experiment, which are shown in rows 7–10 of Table VII. When the number of LCSPM is 1, it can extract fine edge features and fuse the rich features of the backbone network. Therefore, MAE and MSE have a certain increase. The $L2$ layer fused by LCSPM-3 contains background information and shallow features when the number of LCSPM is 3, which leads to a poor counting effect. Thus, the overall performance is optimal when the number of LCSPMs is 2.

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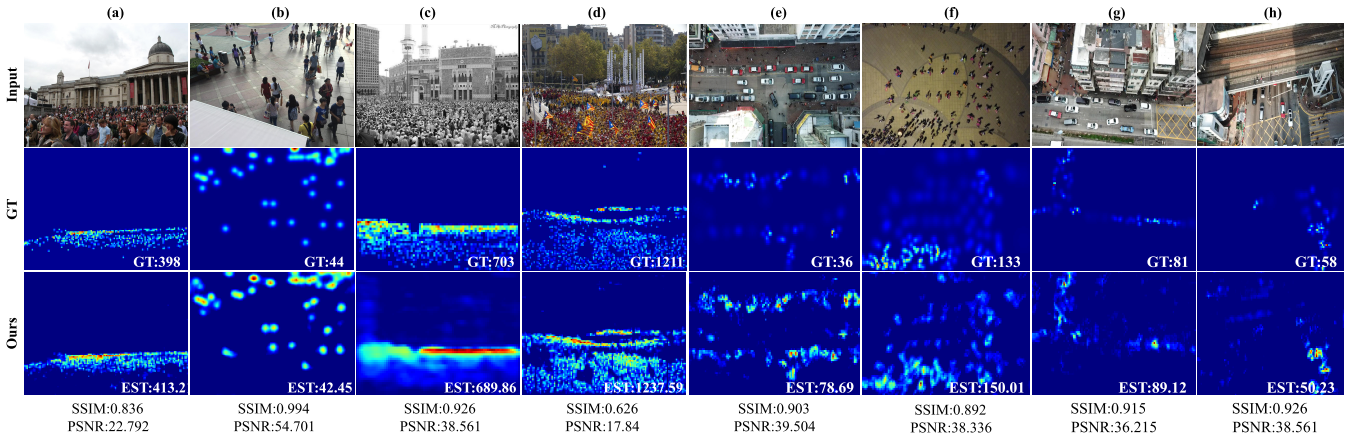


Fig. 6. Visualization results of LMSFFNet on (a) ShanghaiTech Part_A, (b) ShanghaiTech Part_B, (c) UCF_CC_50, (d) UCF-QNRF, and (e)–(h) DroneCrowd datasets (cloudy, night, sparse, and sunny), respectively. The first row represents the original images, the second row represents the GT, and the third row represents the predictions of LMSFFNet.

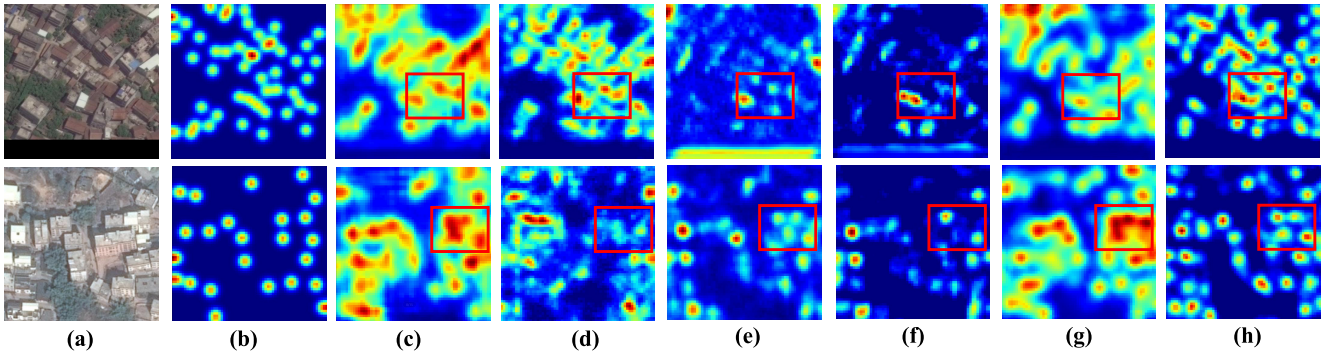


Fig. 7. Density maps generated from each stage of our proposed method. (a) Input images. (b) GT. (c)–(h) Mean density maps generated from baseline, baseline + CA + SA, baseline + CA + SA + LCSPM, baseline + CA + SA + LMCFM + LCSPM, baseline + CA + SA + LCSPM + DCM, and LMSFFNet, respectively.

TABLE IX
ABLATION EXPERIMENT ON BACKBONE STRUCTURE

Methods	Building		Small vehicle		Large vehicle		Ship	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
ResNet-50* [66]	9.36	13.91	324.22	1079.52	16.91	28.52	69.18	100.73
VGGNet-16 [8]	7.93	11.82	238.46	625.90	12.94	20.25	50.45	65.24
MobileNetV2 [16]	7.72	11.90	316.02	598.58	18.55	30.05	73.19	97.13
MobileViT (Ours)	**6.52**	**9.48**	**141.7**	**273**	**12.74**	**27.13**	**49.47**	**70.0**

TABLE X
IMPACTS OF DIFFERENT λ ON RSOC SMALL-VEHICLE

λ	Small-Vehicle	
	MAE	MSE
10	501.05	1301.5
1	151.58	318.51
0	171.09	312.53
0.1	207.66	444.63
0.01(ours)	**141.72**	**273.24**
0.001	177.41	333.89

6) Effects of LMCFM, CA + SA, LCSPM, and DCM: To verify the contribution of each module in LMSFFNet, ablation experiments are conducted on all the categories of RSOC. The experimental results are shown in Table XI. The baseline structure is superior to various complex networks, such as CSRNet, SFCN, and ASPDNet as shown in Table III and Fig. 7(d). It is obvious that the baseline method is disturbed by background suppression and cannot accurately localize the target object.

To address these issues, CA and SA are introduced into our method. After adding an attention mechanism to the baseline, the network can better focus on the objects. Fig. 7(e) shows that the LMCFM module enhances the learning of multi-scale features with fewer parameters and further strengthens the background suppression activity. As shown in Fig. 7(f), the inclusion of attention and DCM enables the network to better perceive information about surrounding objects. However, as shown in Fig. 7(e) and (f), when the LMCFM and DCM modules are used alone, the density map cannot

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TABLE XI
ABLATION STUDIES ON THE RSOC DATASET

Methods					Building		Small_vehicle		Large_vehicle		RSOC_Ship	
Baseline	CA+SA	LCASPP	LMCFM	DCM	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
✓	-	-	-	-	12.16	16.44	250.18	537.43	16.9	36.29	67.65	93.19
✓	✓	-	-	-	9.09	13.49	245.75	646.55	16.72	29.02	63.68	97.81
✓	✓	✓	-	-	7.28	10.61	216.0	451.0	**14.73**	**23.60**	54.48	83.41
✓	✓	✓	-	✓	7.39	10.71	185	334	15.95	29.26	69.52	85.65
✓	✓	✓	✓	-	7.73	10.82	184.43	408.78	20.53	41.6	63.93	87.78
✓	✓	✓	✓	✓	**6.52**	**9.67**	**141.7**	**273.0**	17.34	40.24	**49.47**	**70.0**

TABLE XII
COMPARISON OF THE AVERAGE SQUARE ERROR (MSE), PARAMETERS, FLOPS, AND RUNNING TIME OF OUR METHOD AND THE FOUR MOST ADVANCED METHODS ON THE RSOC DATASET. THE INFERENCE TIMES ARE MEASURED ON INTEL I7-10700K AND NVIDIA RTX TITAN GPU. THE UNIT OF PARAMETERS IS MILLIONS (M), THE UNIT OF FLOPS IS GIGA (G), THE RUNNING TIME OF GPU IS MILLISECONDS (ms), AND THE RUNNING TIME OF CPU IS SECONDS (s) [34]

Methods	Venue	Params(M)	Building (512×512)				Small-Vehicle (768×1024)			
			MAE	GFLOPs	GPU	CPU	MAE	GFLOPs	GPU	CPU
CSRNet [12]	CVPR18	16.26	8.00	108.3	17.2	0.38	443.72	325.3	45.7	20.3
BL [24]	CVPR19	21.50	11.51	108.1	16.3	0.35	168.62	324.1	48.8	20.6
ASPDNet [7]	TGRS20	22.70	7.59	155.3	38	1.2	433.23	465.76	108.9	3.5
PSGCNet [11]	TGRS22	27.51	7.54	128.8	20.5	0.48	157.55	385.80	59.6	1.6
LMSFFNet-XXS	-	**2.13**	7.12	1.87	12.84	0.1	198.14	8.80	24.74	**0.45**
LMSFFNet-XS	-	4.58	**6.52**	**2.93**	**14.2**	**0.34**	**141.7**	**5.6**	**33.2**	**1.9**
LMSFFNet-S	-	9.49	6	13.76	15.36	0.29	205.03	41.30	40.86	1.26

effectively suppress background information. The effect of adding LMCFM and DCM to the network is shown in Fig. 7, and the density map generated by our proposed method is similar to the GT. As a result, the network combines the advantages of each module and achieves excellent performance in the object counting task.

F. Speed Comparison

To further validate the superiority of our method in terms of the numbers of parameters and GFLOPs and the inference speed, we compare our three models (S, XS, and XXS) with advanced methods on RSOC_Building (512 × 512) and RSOC_Small vehicle (1024 × 768), and the experimental results are shown in Table XII. All three models have fewer parameters than the other methods. In addition, the three methods achieve the lowest MAE values on the building dataset. Even the largest model (LMSFFNet-S) has lower inference times on GPU and CPU than PSGCNet [11], ASPDNet [7], BL [23], and CSRNet [12] on the small vehicle datasets. LMSFFNet-XS also achieves the best MAE result, and its MAE value is lower than those of CSRNet and ASPDNet. Meanwhile, XXS obtains the lowest GPU and CPU inference times, taking only 0.45 s per image on CPU. Therefore, our method is also feasible in practical applications.

V. CONCLUSION

In this work, a lightweight multiscale feature fusion network for object counting was proposed to achieve a better balance between counting performance and running speed. Specifically, a MobileViT module as a backbone was selected to reduce the number of network parameters and computing cost in the encoding process. In addition, a cascade structure of CA and SA was introduced to compensate for the weaker feature extraction ability of the lightweight network. In the decoding process, LMCFM as a multiscale feature fusion module was developed to solve the problem that the number of parameters increases with the expansion of the object scale when extracting multiscale features. Furthermore, we use the LCSPM module to mine the subtle features of the target object. We tested LMSFFNet on the RSOC, ShanghaiTech, UCF-QNRF, and UCF_CC_50 datasets and obtained promising performances on remote sensing object counting and crowd counting tasks. The results confirm the effectiveness of our method.

In future research work, we will explore the applicability of this method on various embedded platforms for online public safety assessment and precision transportation and agriculture applications.

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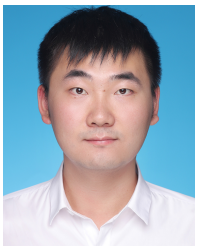
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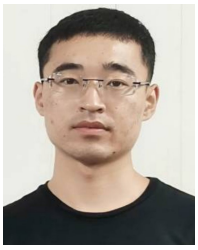
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